Jihun Rho

GPU Performance Engineer
Computer Vision Engineer
B.S. Electronic Engineering, Sejong University
CUDA · Computer Vision
Real-Time Systems

CONTACT

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GitHub profile

Portfolio website

CORE SKILLS

CUDA / GPU
C++, CUDA kernels, memory tuning, stream overlap, occupancy

Profiling
Nsight Systems, Nsight Compute, performance analysis

Computer Vision
OpenCV, image processing, detection, tracking

Inference
TensorRT, YOLO, camera-radar fusion, real-time evaluation

Systems
C/C++, Multithreading, Linux, Windows

Robotics
Robot control, REST, TCP/IP, Modbus, Digital I/O

EDUCATION

B.S. in Electronic Engineering — Sejong University | 2018
ABEEK-accredited engineering program

PROFILE

GPU Performance Engineer | Computer Vision Engineer

GPU Performance Engineer with **4 years and 8 months of hands-on CUDA/C++ experience**, specializing in GPU kernel optimization, high-performance image processing, and industrial machine vision. Bringing 7 years and 6 months of combined industry and research experience spanning computer vision, deep learning, and robotics.
Portfolio: **931njhthe-star.github.io**

EXPERIENCE

Research Engineer — High-Throughput Image Processing Pipeline Development & Optimization

Enclony — Advanced Technology Research Institute | Mar 2021 – Oct 2025

- Developed and optimized high-performance CUDA/C++ machine-vision software for industrial pharmaceutical inspection systems.
- Redesigned image-processing pipelines with selective CUDA acceleration for compute-intensive workloads, retaining efficient CPU processing for lighter tasks and optimizing CPU–GPU workload distribution.
- Achieved **11.2×** higher image-processing throughput than a single-core CPU baseline and approximately **40%** higher throughput than a multi-core CPU-only implementation, reaching **2.52 million images per hour**.
- Processed 10 RGB images (640×480) and 2 3D camera-derived height maps per object for multi-view surface inspection.

Researcher — Deep Learning & Computer Vision

Sejong University — Intelligent Vehicle Recognition Systems Laboratory | Nov 2019 – Oct 2020

- **Vehicle Detection:** Evaluated YOLOv3/v4 and MobileNet-SSD on rooftop-captured video footage. Curated tens to hundreds of thousands of extracted frames, managed JSON/text annotations, and performed detection evaluation and error analysis.
- **Traffic Sign Recognition:** Developed a 3D-to-2D projection simulation using camera geometry and coordinate transformations. Evaluated LeNet and SVM recognition performance for traffic signs rotated around different 3D axes, analyzing the effect of rotation angles on accuracy.

Software Engineer — Robotics & Vision Systems

INA Oriental Motor — Technology Research Center | Jan 2018 – Oct 2019

- Planned and integrated an industrial robotics demonstration system combining autonomous mobile robots, six-axis robots, 3D vision systems, and pneumatic grippers.
- Developed integration software using Python, REST APIs, TCP/IP, Modbus, and digital I/O, implementing monitoring, motion coordination, and gripper-control workflows across devices.
- Integrated deep-learning-based 3D object recognition and random bin-picking into robotics demonstrations.

PROFESSIONAL DEVELOPMENT

Multi-Agent AI Orchestration Program
SK Networks Family × Encore AI Campus | Aug 2026 – Jan 2027 (Expected)

EARLIER TRAINING (2016–2019)

Practical Machine Learning · Python Web Programming · Arduino Electronics · Security & Network Administration

SELECTED PROJECTS

Applied work across perception, robotics integration, and AI systems.

DriveFusion — Real-Time Camera–Radar Perception & Evaluation

View project page | View incident page

Independently built a CARLA-based system combining **YOLO11n inference through a C++/CUDA TensorRT module, vehicle tracking, radar-assisted distance estimation**, and a Qt monitoring dashboard. Across nine weather and time-of-day test conditions (16,491 common samples), radar-assisted tracking reduced overall distance MAE by 5.41%; the reduction reached 39.42% on 437 radar-corrected samples.

Extended the same system with collision video, vehicle reconstruction, sensor timelines, and LLM summaries grounded in selected frames and sensor records.

Technologies: C++, CUDA, TensorRT, YOLO11n, Qt, CARLA, radar fusion, LLM incident review

Robotics & 3D Bin-Picking System Integration

View project page

A professional 3D bin-picking integration project connecting a MiR AMR, Doosan six-axis robot, and Solomon 3D vision, presented through publicly available demonstration materials.

Technologies: Robot control, 3D vision, REST API, TCP/IP, Modbus, digital I/O

ADDITIONAL PROJECTS

SubSync — YouTube English-Learning Extension — View GitHub repository

Built an AI service with FastAPI, Supabase, Redis, and monitoring.

CareerLens — AI Resume & Job Matching — View GitHub repository

Designed a parallel LangGraph pipeline and integrated the service backend.

MOA — AI Service Auto-Healing — View project page

Owned auto-healing in a team project: implemented and tested detection, recovery, and verification flows for six failure scenarios.

Speak Up — English Academy Website — Visit live website

Built and deployed an academy website independently; now in active use.

Experience: AI Agents · Frontend · Backend · Database · Deployment & Operations

SELECTED AWARDS

Excellence Award — 2nd ICT CoC Hackathon (2020) | NIPA & Seoul Metropolitan Government

Awarded for developing a YOLOv3 aerial detection system for unauthorized buildings and rooftop gardens.

Technologies: C++, YOLOv3, deep learning, computer vision, object detection

Additional Awards (2016–2017)

Bronze Award, Sejong University Creative Design Competition (2017) · 4th Place, Sejong University Software Competition (2016) · Outstanding Talent Award (2016)